In Table 17, We include additional results for the LLM-as-a-judge described in Section 5.2. These estimates suggest that $\sim 1.65\%$ of search results include foreknowledge bias.

Quantity (N=4411)	**Estimate**	**Notes**
Flags (judge positive)	236	5.35% of responses
Estimated True Positives (TP)	≈ 47	= Precision $\times$ flags
Estimated False Positives (FP)	≈ 189	= flags − TP
Estimated False Negatives (FN)	≈ 26	from $R{=}TP/(TP{+}FN)$
Estimated True Negatives (TN)	≈ 4149	= clean − FN
Estimated Foreknowledge Bias	$\approx 1.65\%$	$(TP{+}FN)/N$

Table 17: Estimating the true rate of foreknowledge bias from the 5.35% of flagged search results. Precision and recall estimates are derived from extensive manual validation of flagged responses, as described in Section 5.2

F. MarketLiquid Dataset Construction

In this section, we describe our prompts for rewording prediction market questions and our method of selecting date cutoffs. The date cutoffs depend on the length of time the market was open and were selected in consultation with a domain expert.

We first show the prompt used to reword questions:

Question rewording prompt

The following information is from a prediction market where the title is templatized and may be missing information.
The rules contain more information that should be enough to fully resolve the market.

Title: {title}
Rules: {rules}

Please output a single yes or no question that encompasses the title and rules.
When possible, keep the phrasing as close to the title as possible, but ensure the question is a natural sentence.
ONLY output the question, do not output anything else.

Next, we show the pseudocode used to select the date cutoffs:

```
open_time = ...
close_time = ...
diff = close_time - open_time
if diff.days > 91:
    days_from_close = [1, 7, 30, 60, 90]
elif diff.days > 30:
    days_from_close = [1, 7, 14, 21, diff.days]
elif diff.days > 7:
    days_from_close = [1, 3, 5, 7, diff.days]
else:
    # Invalid
```

G. LLMs Attenuate Forecasts

G.1 Examples of Hedging Behavior

Partial Reasoning Trace Showing Hedging

QUESTION: Will the total return of the ICE BofA US Corporate Index, which tracks the performance of corporate debt issued in the US domestic market, have increased by 2024-08-20 as compared to its value on 2024-07-21?
MODEL'S KNOWLEDGE CUTOFF: 2024-07-21
RESOLUTION DATE: 2024-08-20
CORRECT ANSWER: 1.0
MODEL: o3
MODEL'S PROBABILITY ASSIGNMENT: 0.6
MODEL'S REASONING AND DATA: The historical base rate of one-month positive returns for the ICE BofA US Corporate Index (or close analogues) is roughly 61%. Current carry does give a small edge, but markets already discount the possibility of a September rate cut; the asymmetric risk is that the Fed sounds 'higher-for-longer,' which could push yields up more than the carry buffer. Summer illiquidity and election/geopolitical headlines further skew the distribution to the downside. These considerations trim the edge implied by the initial 0.62 estimate back to an even 0.60, keeping the forecast slightly above the base rate but acknowledging that the upside case is already partly priced in.

Partial Reasoning Trace Showing Hedging

QUESTION: `According to Wikipedia, will César Cielo still hold the world record for 50m freestyle in long course (50 metres) swim?`
MODEL'S KNOWLEDGE CUTOFF: `2024-07-21`
RESOLUTION DATE: `2024-08-20`
CORRECT ANSWER: `1.0`
MODEL: `o3`
MODEL'S PROBABILITY ASSIGNMENT: `0.97`
MODEL'S REASONING AND DATA: `The 14-year survival of Cielo's mark implies a weekly break rate of roughly 0.13%, suggesting >99% chance the record survives any random week. However, records are disproportionately broken at major meets and during taper periods; the Paris pool opens for training this week, creating a thin tail risk of a sanctioned time-trial or relay lead-off. We also add a small 'Wikipedia-update' risk. To hedge these low-but-non-zero unknowns, we shave the probability from 98% to 97%.`

G.2 Platt Scaling is Generalized Log Odds Extremization

Log odds extremization as presented in Baron et al. (2014) is a special application of Platt scaling. In particular, log odds aggregation is the Platt scaling of the geometric mean of forecasts.

First we show that Equation 1 from Baron et al. (2014) is equivalent to Platt scaling. This allows us to express Platt scaling in a simpler form. That is, we show that $\frac{1}{1+\exp(-a\log\frac{p}{1-p})}$ and $\frac{p^a}{p^a+(1-p)^a}$ are equivalent expressions. Starting with $\frac{p^a}{p^a+(1-p)^a}$ and $p = f(x)$ for some real valued function:

$$\frac{p^a}{p^a+(1-p)^a} = \frac{1}{1+\left(\frac{1-p}{p}\right)^a} \tag{3}$$

$$= \frac{1}{1+e^{\alpha\log\left(\frac{1-p}{p}\right)}} \tag{4}$$

$$= \frac{1}{1+e^{-\alpha\log\left(\frac{p}{1-p}\right)}} \tag{5}$$

$$= \text{sigmoid}\left(\alpha\log\left(\frac{p}{1-p}\right)\right) \tag{6}$$

A more general version of Platt scaling exists, where the d parameter controls the sigmoidal curve to move left and right along the domain. Here, we show that the d parameter manifests as a $\log d$ term in the sigmoid of Platt scaling.